

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5– Naturalization (BL5,BL6)	Assessment:	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.		
Student Name:	Sairaj C	Reg. No.:	192411018
Section / Batch:	2024	Date:	25-08-26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Apply backtracking techniques systematically for combinatorial problem solving.
- Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required.
- Provide proper justification, state-space representation, and complexity analysis for all solutions.
- Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP-Complete.	Q1

SECTION A – CONSTRAINT-BASED PROBLEM SOLVING

Language: Python C C++ Java

QUESTIONS

Q1

Longest Path Problem (NP-Hard Problem)

100 marks

1/1 answered

Q1 Longest Path Problem (NP-Hard Problem)

100 marks

Find the longest simple path between two vertices in a graph. Clearly define constraints such as vertex non-repetition and path continuity. Analyze how constraints affect solution feasibility. Develop a backtracking-based approach to explore paths. Apply pruning to eliminate shorter paths. Justify correctness and explain NP-Hard classification.

Your Code

```
1 def longest_simple_path(graph, s, t):
2     """
3     Returns (best_length, best_path)
4     Length is measured as number of vertices in the path.
5     graph: adjacency list (dict or list of lists)
6     s, t: source and target vertices
7     """
8     n = len(graph) if not isinstance(graph, dict) else max(graph.keys())
9     visited = [False] * n
10    visited[s] = True
11    best_len = -1
12    best_path = []
13    path = [s]
```

Input

stdin input

Output

Best length (vertices): 4
Best path: [0, 1, 2, 3]

Code of Conduct

I Sairaj C(192411018) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sairaj C

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%				
Constraint Analysis	25%				
Solution Development	25%				
Application of Concepts	15%				
Justification	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____